

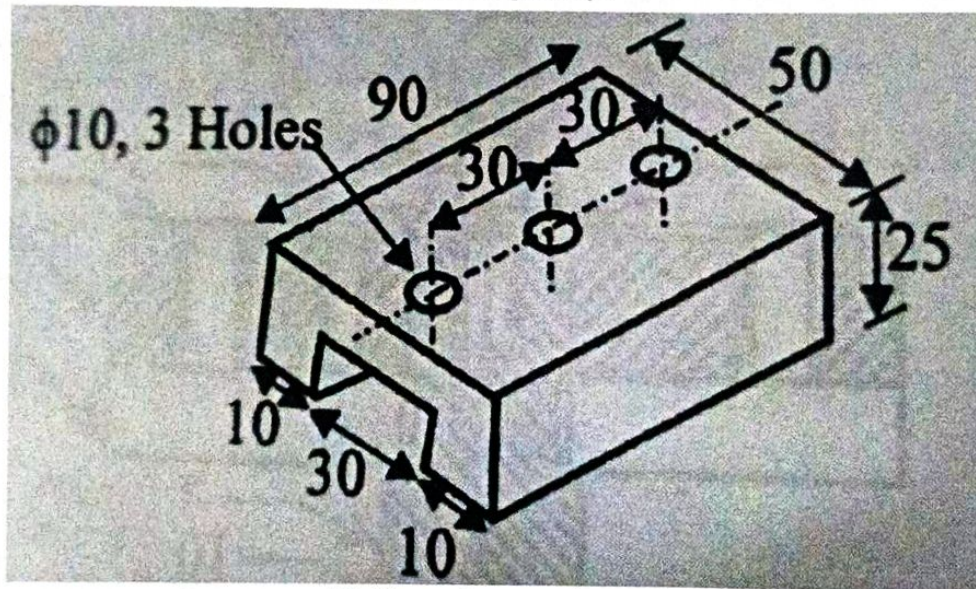
	S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded. In Sem Evaluation-II 2025-26 Class: Final Year (Mechanical Engineering)	
Date: 14/10/2025 Time: 14:30PM To 15:15PM	Subject: Tool Design (PCC-ME401)	Max. Marks: 20

Use of non-programmable calculator is permitted.

Assume suitable data if necessary.

PSG Design Data Book with 3 signatures on each page is allowed.

S.N.	Questions	Marks	CO	BT
Q.1.	Design a preferred drilling jig for use when drilling the ϕ 10mm holes in the component shown in the following fig. with center-to-center distance of 30 mm. Assume that the base has been machined. a) Name the preferred jig. b) Design only bush, locator, clamping device and jig body with sketch. c) Specify appropriate fits and tolerance for critical parts. d) Give a neat parts list with appropriate quantity.	(15)	CO	1,3,4
			1,2	



All dimensions are in mm.

Q.2.	A) List out the four essential features of a milling fixture. State important function (only one) of tenon in fixture. B) Imagine you are tasked with drilling 20 identical holes on 100 aluminum brackets with positional accuracy. How would a jig help in this process, and what type of jig would you choose?	(03)	CO2	1
		(02)	CO2	2

Course Outcomes:

CO 1 Draw and represent the design of mechanical components

CO 2 Selects proper Jig, fixture, positioning, and handling of component to provide ease in manufacture.

CO 3 Demonstrates the sheet metal operation and can explain phenomenon of metal flow during operation.

CO 4 Design and select proper die and tools required for manufacturing.

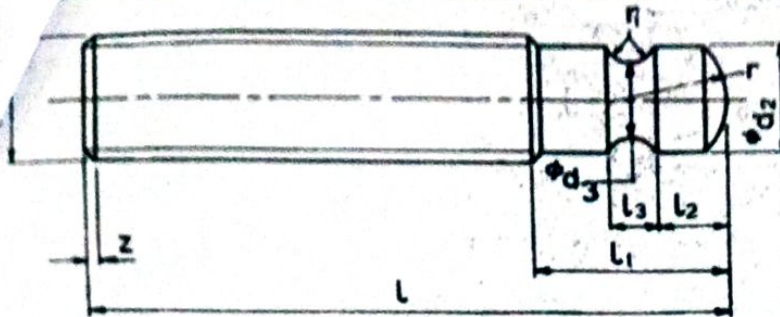
Bloom's Taxonomy (BT):

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

~ALL THE BEST~

"Exams don't judge your worth — they judge your preparation. So, if you're scared, maybe it's not the exam you fear, but your own laziness catching up."

NS FOR PRESSURE OR THRUST PADS IS : 6336

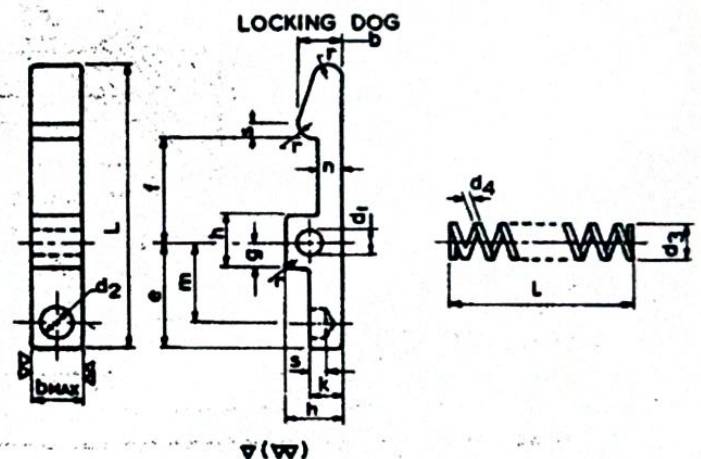


Material — Surface hardening steel—Hardness of the thrust surface ≈ 60 Rc

All dimensions in mm

d_1	d_2	d_3	r	r_1	l_1	l_2	l_3	l	z
M6	4.5	3	3	0.6	6.5	2.2	2.2	30, 50	1.0
M8	6	4	5	0.8	8.5	2.8	3	40, 60	1.25
M10	7	5	6	0.8	10.0	3.2	3	60, 80	1.50
M12	9	6	6	1.0	13.0	4.3	4.5	60, 80, 100	1.75
M16	12	9	9	1.0	17.0	6.3	4.5	80, 100, 125	2.0
M20	15	10	13	1.5	21	7.4	6	100, 125, 150	2.5

LOCKING DOGS : IS : 4295



Material : C 14 for locking dogs

Spring Steel (IS : 727 — 1964) for Springs

All dimensions in mm

Length		Locking dog											Spring (10 Effective Coils)		
L	b	d_1 E 9	d_2	e	f -0.2	g	h	s	k	m	n	r	d_3	d_4	Free-length
45	8	4	5.5	15	15	4	9.5	2	5.5	11	4	1.6	4.6	0.6	16
60	10	5	7.0	20	20	5	12.0	3	7.0	15	5	2.5	5.8	0.8	18
80	14	6	9.0	30	30	6	15.0	5	9.0	23	7	4.0	7.4	1.0	22